

BARABOO WISCONSIN DELLS, WI, USA

WMO: 726503

Lat: 43.522N

Lon: 89.774W

Elev: 298

StdP: 97.80

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.1	-19.1	-27.0	0.3	-21.0	-24.0	0.4	-18.2	9.5	-4.9	8.7	-4.8	1.8	280	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.6	23.2	30.0	22.1	28.2	21.3	25.2	29.0	24.0	27.8	22.9	26.7	4.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	19.5	27.2	22.7	18.1	26.2	22.0	17.3	25.6	78.7	28.4	73.5	27.6	69.4	26.4	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	8.0	7.2	-26.2	34.1	3.0	1.9	-28.4	35.5	-30.2	36.6	-31.9	37.7	-34.2	39.1
			-26.4	26.4	2.9	1.8	-28.5	27.7	-30.2	28.7	-31.8	29.8	-33.9	31.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-7.0	-5.2	1.2	8.0	14.3	19.5	21.9	20.7	16.8	9.8	2.7	-4.2
	DBStd	11.40	6.73	6.60	6.58	5.39	4.78	3.70	3.18	3.01	4.51	5.17	5.60	6.34
	HDD10.0	2085	527	425	284	98	13	0	0	0	3	69	226	439
	HDD18.3	3993	785	658	532	311	143	29	6	11	81	271	468	698
	CDD10.0	1462	0	1	12	39	147	285	369	332	208	61	8	1
	CDD18.3	328	0	0	1	2	19	64	117	84	36	4	0	0
	CDH23.3	2848	0	0	4	30	185	574	1062	656	304	33	0	0
	CDH26.7	788	0	0	0	5	43	154	338	167	77	3	0	0
Wind	WSAvg	3.0	3.1	3.2	3.3	3.7	3.3	2.8	2.4	2.2	2.6	3.1	3.3	3.1
Precipitation	PrecAvg	922	32	31	54	96	107	143	112	114	83	60	52	40
	PrecMax	1185	71	73	147	184	286	342	279	347	282	125	121	94
	PrecMin	662	9	1	11	13	42	16	35	35	10	21	8	6
	PrecStd	150	16	17	32	44	49	78	62	66	57	32	32	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	13.4	22.8	27.1	30.5	32.4	34.9	32.5	31.8	27.2	20.2	13.7
		MCWB	5.5	9.7	15.6	16.4	20.2	22.8	24.6	23.6	22.5	20.7	14.1	12.3
	2%	DB	6.1	8.9	17.7	22.8	27.6	30.2	32.1	30.2	28.8	23.7	17.0	8.9
		MCWB	3.3	5.8	12.3	14.8	18.9	21.7	24.5	22.5	21.2	17.9	12.4	6.8
	5%	DB	2.9	6.2	14.0	20.2	25.3	28.0	30.0	28.2	27.0	21.2	13.8	6.2
		MCWB	1.4	4.0	9.6	12.5	17.1	20.9	22.7	21.5	20.1	16.5	10.2	4.0
	10%	DB	1.6	3.4	11.0	17.4	22.8	27.0	28.1	27.1	24.8	18.0	11.3	3.7
		MCWB	0.7	1.6	7.0	10.8	16.2	20.2	21.9	20.8	18.5	13.8	7.9	2.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	9.8	16.9	18.5	23.0	25.3	28.6	25.9	24.1	22.7	15.5	12.6
		MCDB	7.9	12.5	21.9	24.4	28.0	29.6	31.6	30.1	28.9	26.1	19.0	13.6
	2%	WB	3.4	6.3	14.1	15.9	20.5	23.9	26.0	24.4	22.6	19.4	13.2	7.4
		MCDB	5.6	8.4	17.0	21.4	25.4	28.1	29.3	28.2	26.8	22.0	15.9	8.1
	5%	WB	1.9	4.1	10.2	13.9	19.0	22.7	24.4	23.2	21.3	16.7	10.7	4.2
		MCDB	3.0	6.3	14.0	18.7	23.8	26.9	28.0	26.7	25.1	19.5	13.3	6.0
	10%	WB	0.7	2.0	7.0	11.7	17.5	21.2	23.2	22.1	19.9	14.7	7.9	2.3
		MCDB	1.6	3.5	10.2	16.2	21.3	25.3	26.9	25.4	23.5	18.1	10.5	3.7
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.6	9.5	10.9	11.3	10.9	11.2	10.8	11.6	10.1	8.2	7.4
		MCDBR	8.3	9.6	12.8	15.1	13.7	12.6	12.3	12.1	13.1	13.1	11.9	9.3
		MCWBR	6.8	7.2	8.0	8.5	7.0	6.2	5.8	5.8	6.5	7.8	8.3	7.4
	5% WB	MCDBR	7.6	9.2	11.5	13.0	11.6	11.0	10.5	10.2	11.2	11.0	10.9	8.5
		MCWBR	6.4	7.0	8.0	8.5	6.9	6.5	6.5	6.0	6.6	8.0	8.7	7.4
Clear-Sky Solar Irradiance	taub	0.270	0.311	0.327	0.361	0.390	0.405	0.405	0.398	0.363	0.326	0.291	0.269	
	taud	2.383	2.262	2.377	2.335	2.302	2.294	2.297	2.312	2.401	2.501	2.525	2.447	
	Ebn at Noon	870	877	913	903	882	867	862	856	863	859	844	840	
	Edh at Noon	76	103	106	120	128	129	127	121	103	82	67	66	
All-Sky Solar Radiation	RadAvg	1.72	2.79	3.85	4.51	5.37	5.96	6.07	5.23	4.24	2.73	1.78	1.29	
	RadStd	0.21	0.26	0.46	0.41	0.41	0.43	0.30	0.35	0.34	0.33	0.16	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		-0.91	N/A	N/A	N/A	-1.17	-1.40	N/A	N/A	N/A	N/A
Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page